

Sum of Four Squares

Time limit: 1.00 s

Memory limit: 512 MB

A well known result in number theory is that every non-negative integer can be represented as the sum of four squares of non-negative integers.

You are given a non-negative integer n . Your task is to find four non-negative integers a, b, c and d such that $n = a^2 + b^2 + c^2 + d^2$.

Input

The first line has an integer t : the number of test cases.

Each of the next t lines has an integer n .

Output

For each test case, print four non-negative integers a, b, c and d that satisfy $n = a^2 + b^2 + c^2 + d^2$.

Constraints

- $1 \leq t \leq 1000$
- $0 \leq n \leq 10^7$
- the sum of all n is at most 10^7

Example

Input	Output
3 5 30 322266	2 1 0 0 1 2 3 4 314 159 265 358